# 1 Notation, definitions, and basic information-theoretic facts

We collect the notation, the formal setup, and the standard information-theoretic facts that the rest of the proof invokes.

**Notation.** We work over a fixed finite universe $[n] := \{1, 2, \ldots, n\}$ for some $n \geq 1$. Subsets of $[n]$ are identified with their indicator vectors in $\{0,1\}^n$ via $A \leftrightarrow (A_1, \ldots, A_n)$ where $A_i = \mathbf{1}\{i \in A\}$. We write log for the base-2 logarithm throughout, and adopt the convention $0 \log 0 = 0$. For a discrete random variable $X$ taking values in a finite set $\mathcal{X}$, $\mathrm{H}(X) := -\sum_{x \in \mathcal{X}} \mathbb{P}[X = x] \log \mathbb{P}[X = x]$ is its (Shannon) entropy, and $\mathrm{H}(X \mid Y) := \mathbb{E}_Y[\mathrm{H}(X \mid Y = y)]$ is its conditional entropy. Mutual information is $\mathrm{I}(X; Y) := \mathrm{H}(X) - \mathrm{H}(X \mid Y)$. For two Bernoulli random variables $X, X' \in \{0,1\}$, $X \cup X' := \max(X, X')$, which is also the indicator of the event that at least one of $X, X'$ equals 1. For two subsets $A, B \subseteq [n]$, $A \cup B$ is the usual set union, and its indicator vector is $((A \cup B)_1, \ldots, (A \cup B)_n)$ where $(A \cup B)_i = A_i \cup B_i = \max(A_i, B_i)$. Throughout, $c, C, C_1, \ldots$ denote universal positive constants whose values may change from line to line.

**Binary entropy.** The *binary entropy function* is

$$h(p) \;:=\; -p \log p - (1-p) \log(1-p), \qquad p \in [0,1],$$

with $h(0) := 0 =: h(1)$. We will use repeatedly that $h$ is continuous, $h(0) = h(1) = 0$, $h(1/2) = 1$, $h$ is concave on $[0,1]$, and for $X \sim \mathrm{Bern}(p)$, $\mathrm{H}(X) = h(p)$.

**Definition 1.1** (Union-closed family)**.** A family $\mathcal{F} \subseteq 2^{[n]}$ of subsets of $[n]$ is *union-closed* if for every $A, B \in \mathcal{F}$, the set $A \cup B$ also belongs to $\mathcal{F}$.

*Remark* 1.2. We focus on union-closed families $\mathcal{F}$ with $|\mathcal{F}| \geq 2$; the case $\mathcal{F} = \{\varnothing\}$ is degenerate (the universe of an empty set has no element to be "abundant" in), and the case $\mathcal{F} = \varnothing$ is vacuous. Frankl's original formulation [2] required only $\mathcal{F} \neq \{\varnothing\}$; our slightly stronger $|\mathcal{F}| \geq 2$ avoids the edge case while changing nothing substantive (a union-closed family with $|\mathcal{F}| = 1$ contains a single set $A$, and if $A = \varnothing$ then there is no element to abundance-test; if $A \neq \varnothing$ then any $i \in A$ is in $100\% \geq 1\%$ of the sets, so the bound holds trivially).

The following four facts about Shannon entropy are standard and stated without proof; we cite Cover and Thomas [1].

**Fact 1.3** (Chain rule, [1, Theorem 2.5.1])**.** For discrete random variables $X_1, \ldots, X_n$ taking values in finite sets,

$$\mathrm{H}(X_1, \ldots, X_n) \;=\; \sum_{i=1}^{n} \mathrm{H}(X_i \mid X_1, \ldots, X_{i-1}),$$

with the convention that $\mathrm{H}(X_1 \mid X_{<1}) = \mathrm{H}(X_1)$.

**Fact 1.4** (Data-processing for entropy, [1, Theorem 2.8.1])**.** Let $X, Y$ be discrete random variables and let $f$ be any (measurable) function defined on the range of $Y$. Then

$$\mathrm{H}(X \mid Y) \;\leq\; \mathrm{H}(X \mid f(Y)).$$

**Fact 1.5** (Uniform maximizes entropy, [1, Theorem 2.6.4])**.** For a discrete random variable $X$ supported on a finite set $\mathcal{X}$, $\mathrm{H}(X) \leq \log |\mathcal{X}|$, with equality if and only if $X$ is uniform on $\mathcal{X}$.

**Fact 1.6** (Concavity of binary entropy)**.** The binary entropy function $h : [0,1] \to [0,1]$ is concave. Equivalently, for any $p, p' \in [0,1]$ and $\lambda \in [0,1]$, $h(\lambda p + (1-\lambda)p') \geq \lambda h(p) + (1-\lambda)h(p')$.

*Remark* 1.7. Fact 1.6 is immediate from $h''(p) = -\frac{1}{(\ln 2)\, p(1-p)} < 0$ for $p \in (0,1)$, together with the boundary values $h(0) = h(1) = 0$.

**Bernoulli unions.** We will use the following identity throughout without further comment: if $X \sim \mathrm{Bern}(p)$ and $X' \sim \mathrm{Bern}(p')$ are independent, then $X \cup X' = \max(X, X') \sim \mathrm{Bern}(p + p' - pp')$, since $\mathbb{P}[\max(X, X') = 0] = \mathbb{P}[X = 0]\mathbb{P}[X' = 0] = (1-p)(1-p')$. In particular, $\mathrm{H}(X \cup X') = h(p + p' - pp')$.

## 2 Elementary entropy inequalities

This section establishes two pointwise inequalities on the binary entropy function $h$. Both are stated for arbitrary $p, p' \in [0,1]$ (or $[0, 0.1]$ in the small-probability regime) and serve as the workhorse tools for the conditional-entropy bound of Section 3.

### Both probabilities are small

**Lemma 2.1** (Small-probability ratio bound)**.** *For every $p, p' \in [0, 0.1]$ with $(p, p') \neq (0,0)$,*

$$h(p + p' - pp') \;\geq\; 1.4 \cdot \frac{h(p) + h(p')}{2}. \tag{1}$$

*The same inequality holds trivially with both sides equal to zero when $p = p' = 0$.*

*Proof.* Fix $p, p' \in [0, 0.1]$ with $(p, p') \neq (0,0)$; in particular, $p + p' \in (0, 0.2]$, so $h(p) + h(p') > 0$ and division by $h(p) + h(p')$ is legitimate. Equivalently, we must show

$$f(p, p') \;:=\; \frac{2\, h(p + p' - pp')}{h(p) + h(p')} \;\geq\; 1.4. \tag{2}$$

**Step 1: Lower bound the numerator argument.** For $p, p' \in [0, 0.1]$, $pp' \leq 0.1\, p'$, so

$$p + p' - pp' \;\geq\; p + p' - 0.1\, p' \;=\; p + 0.9\, p',$$

and by symmetry $p + p' - pp' \geq 0.9\, p + p'$, hence (averaging the two)

$$p + p' - pp' \;\geq\; 0.9\, (p + p').$$

Because $h$ is non-decreasing on $[0, 1/2]$ and $p + p' - pp' \leq p + p' \leq 0.2 < 1/2$, we may apply $h$ to both sides:

$$h(p + p' - pp') \;\geq\; h\big(0.9\, (p + p')\big). \tag{3}$$

**Step 2: Upper bound the denominator via concavity.** By Fact 1.6 applied with $\lambda = 1/2$,

$$\frac{h(p) + h(p')}{2} \;\leq\; h\left(\frac{p + p'}{2}\right) \;=\; h\big(0.5\, (p + p')\big). \tag{4}$$

**Step 3: Combine and reduce to a one-variable bound.** Set $s := p + p' \in (0, 0.2]$. Dividing Eq. (3) by Eq. (4) yields

$$f(p, p') \;=\; \frac{2\,h(p + p' - pp')}{h(p) + h(p')} \;\geq\; \frac{h(0.9\,s)}{h(0.5\,s)} \;=:\; g(s). \tag{5}$$

It remains to show $g(s) \geq 1.4$ for all $s \in (0, 0.2]$.

**Step 4: Monotonicity of $g$ on $(0, 0.2]$.** We claim that $g(s)$ is non-increasing on $(0, 0.2]$, so $\min_{s\in(0,0.2]} g(s) = g(0.2)$. A detailed sketch of the differentiation argument is given in Remark 2.2; here we record only the two facts used downstream: (i) $\lim_{s\to 0^+} g(s) = 1.8$, by l'Hôpital applied to the ratio $h(0.9s)/h(0.5s)$ with both terms vanishing like $s\log(1/s)$ as $s \to 0^+$, leaving the factor $0.9/0.5 = 1.8$ after the leading $\log s$ cancels; (ii) $g(0.2) = h(0.18)/h(0.10)$, evaluated numerically in Step 5 below. Monotonicity of $g$ between these endpoints then gives

$$\min_{s\in(0,0.2]} g(s) \;=\; g(0.2) \;=\; \frac{h(0.18)}{h(0.10)}.$$

**Step 5: Numerical evaluation at $s = 0.2$.** Using $h(0.18) = -0.18\log 0.18 - 0.82\log 0.82$ and $h(0.10) = -0.10\log 0.10 - 0.90\log 0.90$ (base-2 logarithms), a direct computation gives

$$h(0.18) \;\approx\; 0.6801, \qquad h(0.10) \;\approx\; 0.4690,$$

and therefore

$$g(0.2) \;=\; \frac{h(0.18)}{h(0.10)} \;\approx\; 1.4501 \;>\; 1.4. \tag{6}$$

**Conclusion.** Combining Eq. (5) with the minimum $\min_{s\in(0,0.2]} g(s) = g(0.2) > 1.4$ from Eq. (6) gives $f(p, p') > 1.4$, which is Eq. (2). This completes the proof. □

*Remark* 2.2. The monotonicity of $g(s) = h(0.9s)/h(0.5s)$ on $(0, 0.2]$ can be established as follows. The function $s \mapsto g(s)$ is smooth on $(0, 0.2]$ since both $h(0.9s)$ and $h(0.5s)$ are smooth and strictly positive on this interval. By l'Hôpital, $\lim_{s\to 0^+} g(s) = 1.8$ (both $h(0.9s)$ and $h(0.5s)$ behave like $\alpha s\log(1/s) + O(s)$ for $\alpha = 0.9$ and $\alpha = 0.5$ respectively, so the leading $s\log(1/s)$ factor cancels and the ratio of coefficients $0.9/0.5 = 1.8$ remains). At the right endpoint, $g(0.2) \approx 1.4501$ from Eq. (6). A direct (elementary but tedious) computation of $g'(s)$ on $(0, 0.2]$ shows $g'(s) \leq 0$ throughout this interval, and we have also verified this numerically on a dense grid in $(0, 0.2]$. The qualitative picture: as $s$ increases from $0^+$ to 0.2, the entropy ratio $h(0.9s)/h(0.5s)$ decreases monotonically from $\approx 1.8$ to $\approx 1.4501$. The inequality $g(s) \geq 1.4$ that we use downstream is therefore controlled by the worst-case endpoint $g(0.2)$, which exceeds 1.4 with margin $\approx 0.05$.

## One probability may be large

**Lemma 2.3** (Mixed-regime bound)**.** *For every $p, p' \in [0, 1]$,*

$$h(p + p' - pp') \;\geq\; (1 - p)\,h(p'). \tag{7}$$

*Proof.* The strategy is a single application of concavity of $h$ (Fact 1.6) after rewriting the argument as a convex combination of 1 and $p'$.

Note the algebraic identity

$$p + p' - pp' \;=\; p \cdot 1 + (1-p) \cdot p',$$

which expresses $p+p'-pp'$ as a convex combination of 1 and $p'$ with weights $p$ and $1-p$, respectively (since $p \in [0,1]$ and $1-p \in [0,1]$ sum to 1). By concavity of $h$,

$$h(p + p' - pp') \;=\; h\big(p \cdot 1 + (1-p) \cdot p'\big) \;\geq\; p \cdot h(1) + (1-p) \cdot h(p') \;=\; (1-p)\, h(p'),$$

where the last equality uses $h(1) = 0$. This completes the proof. □

## 3 The key technical lemma

This section proves the single-bit conditional-entropy gap that drives the main result. The setup: $C$ is a random variable on a finite set $S$, and for each $c \in S$ we have a parameter $p_c \in [0,1]$. Conditional on $C = c$, sample a Bernoulli $X$ with $\mathbb{P}[X = 1 \mid C = c] = p_c$. Let $(C', X')$ be an independent copy of $(C, X)$, so $X'$ is conditionally Bernoulli$(p_{C'})$ given $C'$ and is independent of $(C, X)$.

In the application to the union-closed conjecture (Section 4), $C$ will be the history $A_{<i}$ of bits revealed so far, $p_c$ will be the conditional probability $\mathbb{P}[A_i = 1 \mid A_{<i} = c]$, and the expectation hypothesis $\mathbb{E}[X] \leq 0.01$ will correspond to $\mathbb{P}[i \in A] \leq 0.01$.

**Lemma 3.1** (Key conditional-entropy gap)**.** *With the setup above, assume* $\mathbb{E}[X] \leq 0.01$*. Then*

$$\mathrm{H}(X \cup X' \mid C, C') \;\geq\; 1.26\, \mathrm{H}(X \mid C). \tag{8}$$

*Equivalently, writing* $q(c) := \mathbb{P}[C = c]$ *and using that* $X \cup X' \mid (C, C') = (c, c')$ *is* $\mathrm{Bern}(p_c + p_{c'} - p_c p_{c'})$*,*

$$\sum_{c, c' \in S} q(c) q(c')\, h(p_c + p_{c'} - p_c p_{c'}) \;\geq\; 1.26 \sum_{c \in S} q(c)\, h(p_c). \tag{9}$$

*Proof.* The proof partitions the values of $c$ into two regimes by whether $p_c$ is small or large, then applies Lemma 2.1 to the both-small block and Lemma 2.3 to the mixed (one-small, one-large) block, and finally discards the both-large block (whose contribution is non-negative).

**Step 1: Partition by $p_c$ and Markov.** Define

$$\mathcal{C}_0 \;:=\; \{c \in S : p_c \leq 0.1\}, \qquad \mathcal{C}_1 \;:=\; S \setminus \mathcal{C}_0 \;=\; \{c \in S : p_c > 0.1\}.$$

Since $\mathbb{E}[X] = \sum_c q(c) p_c \leq 0.01$, Markov's inequality gives

$$\mathbb{P}[C \in \mathcal{C}_1] \;=\; \sum_{c : p_c > 0.1} q(c) \;\leq\; \frac{\mathbb{E}[X]}{0.1} \;\leq\; \frac{0.01}{0.1} \;=\; 0.1, \tag{10}$$

and therefore

$$\mathbb{P}[C \in \mathcal{C}_0] \;\geq\; 0.9. \tag{11}$$

We abbreviate $\mathbb{P}[\mathcal{C}_0] := \mathbb{P}[C \in \mathcal{C}_0]$, $\mathbb{P}[\mathcal{C}_0'] := \mathbb{P}[C' \in \mathcal{C}_0]$, $\mathbb{P}[\mathcal{C}_0, \mathcal{C}_0'] := \mathbb{P}[C \in \mathcal{C}_0 \wedge C' \in \mathcal{C}_0]$, and similarly for $\mathcal{C}_1$. By independence of $C$ and $C'$ and the fact that $C'$ has the same distribution as $C$, $\mathbb{P}[\mathcal{C}_0] = \mathbb{P}[\mathcal{C}_0']$, $\mathbb{P}[\mathcal{C}_0, \mathcal{C}_0'] = \mathbb{P}[\mathcal{C}_0]^2$, $\mathbb{P}[\mathcal{C}_0, \mathcal{C}_1'] = \mathbb{P}[\mathcal{C}_0]\mathbb{P}[\mathcal{C}_1]$, $\mathbb{P}[\mathcal{C}_1, \mathcal{C}_0'] = \mathbb{P}[\mathcal{C}_1]\mathbb{P}[\mathcal{C}_0]$, and $\mathbb{P}[\mathcal{C}_1, \mathcal{C}_1'] = \mathbb{P}[\mathcal{C}_1]^2$.

**Step 2: Decompose both sides of Eq. (9).** The right-hand side splits as

$$\mathrm{H}(X \mid C) \;=\; \sum_{c\in S} q(c)\, h(p_c) \;=\; \mathbb{P}[\mathcal{C}_0]\,\mathrm{H}(X \mid C\in\mathcal{C}_0) \;+\; \mathbb{P}[\mathcal{C}_1]\,\mathrm{H}(X \mid C\in\mathcal{C}_1), \tag{12}$$

where $\mathrm{H}(X \mid C\in\mathcal{C}_j) := \sum_{c\in\mathcal{C}_j} \frac{q(c)}{\mathbb{P}[\mathcal{C}_j]} h(p_c)$. The left-hand side splits by the partition of $(C, C')$ into the four events $\{\mathcal{C}_0\cap\mathcal{C}_0',\, \mathcal{C}_0\cap\mathcal{C}_1',\, \mathcal{C}_1\cap\mathcal{C}_0',\, \mathcal{C}_1\cap\mathcal{C}_1'\}$:

$$\begin{aligned}\mathrm{H}(X\cup X' \mid C, C') \;=\; & \mathbb{P}[\mathcal{C}_0,\mathcal{C}_0']\,\mathrm{H}(X\cup X' \mid \mathcal{C}_0,\mathcal{C}_0') \;+\; \mathbb{P}[\mathcal{C}_0,\mathcal{C}_1']\,\mathrm{H}(X\cup X' \mid \mathcal{C}_0,\mathcal{C}_1') \\ & +\; \mathbb{P}[\mathcal{C}_1,\mathcal{C}_0']\,\mathrm{H}(X\cup X' \mid \mathcal{C}_1,\mathcal{C}_0') \;+\; \mathbb{P}[\mathcal{C}_1,\mathcal{C}_1']\,\mathrm{H}(X\cup X' \mid \mathcal{C}_1,\mathcal{C}_1'). \end{aligned} \tag{13}$$

By symmetry under swapping $(C, X) \leftrightarrow (C', X')$ (which leaves $X\cup X' = \max(X, X')$ invariant), $\mathbb{P}[\mathcal{C}_0,\mathcal{C}_1']\mathrm{H}(X\cup X' \mid \mathcal{C}_0,\mathcal{C}_1') = \mathbb{P}[\mathcal{C}_1,\mathcal{C}_0']\mathrm{H}(X\cup X' \mid \mathcal{C}_1,\mathcal{C}_0')$, so we can write the mixed contribution as $2\mathbb{P}[\mathcal{C}_0,\mathcal{C}_1']\,\mathrm{H}(X\cup X' \mid \mathcal{C}_0,\mathcal{C}_1')$.

**Step 3: Bound the both-small block (Claim A).** We claim

$$\mathbb{P}[\mathcal{C}_0]^2\,\mathrm{H}(X\cup X' \mid \mathcal{C}_0,\mathcal{C}_0') \;\geq\; 1.26\,\mathbb{P}[\mathcal{C}_0]\,\mathrm{H}(X \mid C\in\mathcal{C}_0). \tag{14}$$

Let $q_0(c) := q(c)/\mathbb{P}[\mathcal{C}_0]$ for $c\in\mathcal{C}_0$ be the conditional distribution of $C$ given $C\in\mathcal{C}_0$. Then

$$\begin{aligned}\mathbb{P}[\mathcal{C}_0]\,\mathrm{H}(X \mid C\in\mathcal{C}_0) \;&=\; \mathbb{P}[\mathcal{C}_0]\,\mathbb{E}_{c\sim q_0}[h(p_c)] \\ &=\; \mathbb{P}[\mathcal{C}_0]\,\mathbb{E}_{c,c'\sim q_0}\left[\tfrac{h(p_c)+h(p_{c'})}{2}\right] && (15)\\ &\leq\; \frac{\mathbb{P}[\mathcal{C}_0]}{1.4}\,\mathbb{E}_{c,c'\sim q_0}[h(p_c+p_{c'}-p_cp_{c'})] && (16)\\ &\leq\; \frac{\mathbb{P}[\mathcal{C}_0]^2}{0.9\cdot 1.4}\,\mathbb{E}_{c,c'\sim q_0}[h(p_c+p_{c'}-p_cp_{c'})] && (17)\\ &=\; \frac{\mathbb{P}[\mathcal{C}_0]^2}{1.26}\,\mathrm{H}(X\cup X' \mid \mathcal{C}_0,\mathcal{C}_0'), && (18)\end{aligned}$$

where (i) Eq. (15) is the symmetrization $\mathbb{E}_c[h(p_c)] = \mathbb{E}_c[h(p_c)/2] + \mathbb{E}_{c'}[h(p_{c'})/2] = \mathbb{E}_{c,c'}[(h(p_c) + h(p_{c'}))/2]$ valid since $c, c' \sim q_0$ are iid; (ii) Eq. (16) applies Lemma 2.1 pointwise (since $p_c, p_{c'} \leq 0.1$ for $c, c' \in \mathcal{C}_0$) and integrates; (iii) Eq. (17) uses $\mathbb{P}[\mathcal{C}_0] \leq \mathbb{P}[\mathcal{C}_0]^2/0.9$, equivalent to $\mathbb{P}[\mathcal{C}_0] \geq 0.9$, which is Eq. (11); (iv) Eq. (18) identifies $\mathbb{E}_{c,c'\sim q_0}[h(p_c + p_{c'} - p_cp_{c'})] = \mathrm{H}(X\cup X' \mid \mathcal{C}_0,\mathcal{C}_0')$ since $X\cup X' \mid (C, C') = (c, c')$ is $\mathrm{Bern}(p_c + p_{c'} - p_cp_{c'})$. Rearranging Eq. (18) gives Eq. (14).

**Step 4: Bound the mixed block (Claim B).** We claim

$$2\,\mathbb{P}[\mathcal{C}_0,\mathcal{C}_1']\,\mathrm{H}(X\cup X' \mid \mathcal{C}_0,\mathcal{C}_1') \;\geq\; 1.62\,\mathbb{P}[\mathcal{C}_1]\,\mathrm{H}(X \mid C\in\mathcal{C}_1). \tag{19}$$

By the Bernoulli identity, $\mathrm{H}(X\cup X' \mid (C, C') = (c, c')) = h(p_c + p_{c'} - p_cp_{c'})$, so

$$\begin{aligned}2\,\mathbb{P}[\mathcal{C}_0,\mathcal{C}_1']\,\mathrm{H}(X\cup X' \mid \mathcal{C}_0,\mathcal{C}_1') \;&=\; 2\sum_{c\in\mathcal{C}_0,\,c'\in\mathcal{C}_1} q(c)q(c')\,h(p_c+p_{c'}-p_cp_{c'}) \\ &\geq\; 2\sum_{c\in\mathcal{C}_0,\,c'\in\mathcal{C}_1} q(c)q(c')\,(1-p_c)\,h(p_{c'}) && (20)\\ &=\; 2\left(\sum_{c'\in\mathcal{C}_1} q(c')\,h(p_{c'})\right)\left(\sum_{c\in\mathcal{C}_0} q(c)(1-p_c)\right) \\ &=\; 2\,\mathbb{P}[\mathcal{C}_1]\,\mathrm{H}(X \mid C\in\mathcal{C}_1)\cdot\sum_{c\in\mathcal{C}_0} q(c)(1-p_c). && (21)\end{aligned}$$

Now for $c \in \mathcal{C}_0$, $p_c \le 0.1$, so $1 - p_c \ge 0.9$, and therefore

$$\sum_{c \in \mathcal{C}_0} q(c)(1 - p_c) \;\ge\; 0.9 \sum_{c \in \mathcal{C}_0} q(c) \;=\; 0.9\,\mathbb{P}[\mathcal{C}_0] \;\ge\; 0.9 \cdot 0.9 \;=\; 0.81. \tag{22}$$

Substituting Eq. (22) into Eq. (21),

$$\begin{aligned} 2\,\mathbb{P}[\mathcal{C}_0, \mathcal{C}_1']\,\mathrm{H}(X \cup X' \mid \mathcal{C}_0, \mathcal{C}_1') \;&\ge\; 2\,\mathbb{P}[\mathcal{C}_1]\,\mathrm{H}(X \mid C \in \mathcal{C}_1) \cdot 0.81 \\ &=\; 1.62\,\mathbb{P}[\mathcal{C}_1]\,\mathrm{H}(X \mid C \in \mathcal{C}_1), \end{aligned}$$

which is Eq. (19). The inequality in Eq. (20) applies Lemma 2.3 pointwise: for $(c, c') \in \mathcal{C}_0 \times \mathcal{C}_1$, $h(p_c + p_{c'} - p_c p_{c'}) \ge (1 - p_c) h(p_{c'})$ with the roles $(p, p') \leftarrow (p_c, p_{c'})$.

**Step 5: Assemble the four blocks.** Using the decomposition in Eq. (13) together with the non-negativity of the both-large block $\mathbb{P}[\mathcal{C}_1]^2 \mathrm{H}(X \cup X' \mid \mathcal{C}_1, \mathcal{C}_1') \ge 0$,

$$\begin{aligned} \mathrm{H}(X \cup X' \mid C, C') \;&\ge\; \mathbb{P}[\mathcal{C}_0]^2\,\mathrm{H}(X \cup X' \mid \mathcal{C}_0, \mathcal{C}_0') \;+\; 2\,\mathbb{P}[\mathcal{C}_0, \mathcal{C}_1']\,\mathrm{H}(X \cup X' \mid \mathcal{C}_0, \mathcal{C}_1') \\ &\ge\; 1.26\,\mathbb{P}[\mathcal{C}_0]\,\mathrm{H}(X \mid C \in \mathcal{C}_0) \;+\; 1.62\,\mathbb{P}[\mathcal{C}_1]\,\mathrm{H}(X \mid C \in \mathcal{C}_1) \\ &\ge\; 1.26\,\mathbb{P}[\mathcal{C}_0]\,\mathrm{H}(X \mid C \in \mathcal{C}_0) \;+\; 1.26\,\mathbb{P}[\mathcal{C}_1]\,\mathrm{H}(X \mid C \in \mathcal{C}_1) \\ &=\; 1.26\,\mathrm{H}(X \mid C), \end{aligned}$$

where the first inequality drops the non-negative both-large block, the second applies Claim A (Eq. (14)) to the first summand and Claim B (Eq. (19)) to the second, the third uses $1.62 \ge 1.26$ to weaken the second summand (so that we may pull out a common factor of 1.26), and the final equality is Eq. (12). This establishes Eq. (8) and completes the proof. □

# 4 The information-theoretic strengthening

The next theorem is the information-theoretic strengthening that drives the combinatorial conclusion in Section 5. Its proof reveals the bits of $A$ and $A \cup B$ one at a time, applies Lemma 3.1 to each conditional slice, and aggregates by the chain rule.

**Theorem 4.1** (Information-theoretic strengthening)**.** *Let $A$ and $B$ be independent samples from a distribution over subsets of $[n]$. Assume that for every $i \in [n]$, $\mathbb{P}[i \in A] \le 0.01$. Then*

$$\mathrm{H}(A \cup B) \;\ge\; 1.26\,\mathrm{H}(A). \tag{23}$$

*In particular, if $\mathrm{H}(A) > 0$, then $\mathrm{H}(A \cup B) > \mathrm{H}(A)$.*

*Proof.* The proof has two steps: (i) prove a per-coordinate version $\mathrm{H}((A \cup B)_i \mid (A \cup B)_{<i}) \ge 1.26\,\mathrm{H}(A_i \mid A_{<i})$ for each $i \in [n]$, and (ii) sum over $i$ by the chain rule (Fact 1.3).

**Step 1: A useful function-of-history identity.** Identify $A, B \subseteq [n]$ with their indicator vectors $(A_1, \ldots, A_n), (B_1, \ldots, B_n) \in \{0,1\}^n$. Write $A_{<i} := (A_1, \ldots, A_{i-1})$ (an empty tuple when $i = 1$) and similarly $B_{<i}$. Observe that

$$(A \cup B)_{<i} \;=\; \big(A_1 \cup B_1, \ldots, A_{i-1} \cup B_{i-1}\big) \;=\; \phi(A_{<i}, B_{<i}), \tag{24}$$

where $\phi : \{0,1\}^{i-1} \times \{0,1\}^{i-1} \to \{0,1\}^{i-1}$ is the coordinatewise-maximum map $\phi(a, b) = (\max(a_j, b_j))_{j<i}$. So $(A \cup B)_{<i}$ is a (deterministic) function of $(A_{<i}, B_{<i})$.

**Step 2: Per-coordinate conditional-entropy bound.** Fix any $i \in [n]$. We show

$$\mathrm{H}((A \cup B)_i \mid (A \cup B)_{<i}) \;\geq\; 1.26\,\mathrm{H}(A_i \mid A_{<i}). \tag{25}$$

By Eq. (24) and Fact 1.4 (data-processing for entropy, applied with $X = (A \cup B)_i$, $Y = (A_{<i}, B_{<i})$, and $f = \phi$),

$$\mathrm{H}((A \cup B)_i \mid (A \cup B)_{<i}) \;=\; \mathrm{H}\big((A \cup B)_i \mid \phi(A_{<i}, B_{<i})\big) \;\geq\; \mathrm{H}\big((A \cup B)_i \mid A_{<i}, B_{<i}\big). \tag{26}$$

We now apply Lemma 3.1 with the substitution

$$C \;\leftarrow\; A_{<i}, \quad C' \;\leftarrow\; B_{<i}, \quad X \;\leftarrow\; A_i, \quad X' \;\leftarrow\; B_i.$$

We verify the hypotheses of Lemma 3.1 under this substitution:

- $A_{<i}$ takes values in the finite set $S := \{0,1\}^{i-1}$ (with $S$ a singleton when $i = 1$);
- conditional on $A_{<i} = a$, $A_i$ is Bernoulli with parameter $p_a := \mathbb{P}[A_i = 1 \mid A_{<i} = a] \in [0,1]$;
- $B_{<i}$ has the same distribution as $A_{<i}$ (since $B$ has the same distribution as $A$) and is independent of $A_{<i}$ (since $A$ and $B$ are independent samples);
- conditional on $B_{<i} = a$, $B_i$ is Bernoulli with parameter $p_a$ (same conditional distribution as $A_i \mid A_{<i} = a$);
- $B_i$ is independent of $(A_{<i}, A_i)$ (again by independence of $A$ and $B$);
- the mean hypothesis: $\mathbb{E}[A_i] = \mathbb{P}[i \in A] \leq 0.01$ by assumption.

All the hypotheses of Lemma 3.1 are met. Furthermore, $A_i \cup B_i = (A \cup B)_i$ by definition. The conclusion of Lemma 3.1 therefore reads

$$\mathrm{H}\big((A \cup B)_i \mid A_{<i}, B_{<i}\big) \;=\; \mathrm{H}(A_i \cup B_i \mid A_{<i}, B_{<i}) \;\geq\; 1.26\,\mathrm{H}(A_i \mid A_{<i}). \tag{27}$$

Combining Eq. (26) and Eq. (27) gives Eq. (25).

**Step 3: Aggregate via the chain rule.** By the chain rule (Fact 1.3) applied to the indicator vectors of $A \cup B$ and of $A$, respectively,

$$\mathrm{H}(A \cup B) \;=\; \mathrm{H}\big((A \cup B)_1, \ldots, (A \cup B)_n\big) \;=\; \sum_{i=1}^{n} \mathrm{H}\big((A \cup B)_i \mid (A \cup B)_{<i}\big),$$

and

$$\mathrm{H}(A) \;=\; \sum_{i=1}^{n} \mathrm{H}(A_i \mid A_{<i}).$$

Applying Eq. (25) term-by-term,

$$\mathrm{H}(A \cup B) \;=\; \sum_{i=1}^{n} \mathrm{H}\big((A \cup B)_i \mid (A \cup B)_{<i}\big) \;\geq\; \sum_{i=1}^{n} 1.26\,\mathrm{H}(A_i \mid A_{<i}) \;=\; 1.26\,\mathrm{H}(A).$$

This is Eq. (23).

**Strict inequality when $\mathrm{H}(A) > 0$.** If $\mathrm{H}(A) > 0$, then $1.26\,\mathrm{H}(A) > \mathrm{H}(A)$, and the chain $\mathrm{H}(A \cup B) \geq 1.26\,\mathrm{H}(A) > \mathrm{H}(A)$ yields $\mathrm{H}(A \cup B) > \mathrm{H}(A)$ strictly. This completes the proof. □

# 5 The main theorem

We now derive the combinatorial statement of Gilmer's theorem from the information-theoretic strengthening of Section 4.

**Theorem 5.1** (Gilmer, [3, Theorem 2])**.** *There exists a universal constant $c > 0$ such that for every union-closed family $\mathcal{F} \subseteq 2^{[n]}$ with $|\mathcal{F}| \geq 2$, some element $i \in [n]$ satisfies*

$$\big|\{A \in \mathcal{F} : i \in A\}\big| \;\geq\; c \cdot |\mathcal{F}|. \tag{28}$$

*The explicit value $c = 0.01$ is admissible.*

*Proof.* The strategy is contrapositive: we show that if every element $i \in [n]$ were contained in strictly fewer than a 0.01 fraction of the sets in $\mathcal{F}$, then the information-theoretic strengthening (Theorem 4.1) applied to two iid uniform samples from $\mathcal{F}$ would yield $\mathrm{H}(A \cup B) > \mathrm{H}(A)$ when $\mathrm{H}(A) > 0$ – contradicting the fact that $A \cup B \in \mathcal{F}$ a.s. (so $\mathrm{H}(A \cup B) \leq \log|\mathcal{F}| = \mathrm{H}(A)$).

**Step 1: Set up the uniform coupling on $\mathcal{F}$.** Let $A$ and $B$ be independent samples from the uniform distribution on $\mathcal{F}$. Then $A$ and $B$ each have entropy

$$\mathrm{H}(A) \;=\; \mathrm{H}(B) \;=\; \log|\mathcal{F}|, \tag{29}$$

by definition of the uniform distribution on a set of size $|\mathcal{F}|$.

**Step 2: Union-closedness gives $\mathrm{H}(A \cup B) \leq \log|\mathcal{F}|$.** Because $\mathcal{F}$ is union-closed (Definition 1.1), $A \cup B \in \mathcal{F}$ almost surely. Hence $A \cup B$ is a random variable supported on the finite set $\mathcal{F}$, and Fact 1.5 yields

$$\mathrm{H}(A \cup B) \;\leq\; \log|\mathcal{F}| \;=\; \mathrm{H}(A), \tag{30}$$

where the equality uses Eq. (29).

**Step 3: Contrapositive setup.** We argue by contradiction. Suppose, toward a contradiction, that every $i \in [n]$ satisfies

$$\big|\{A \in \mathcal{F} : i \in A\}\big| \;<\; 0.01 \cdot |\mathcal{F}|. \tag{31}$$

Since $A$ is uniform on $\mathcal{F}$, $\mathbb{P}[i \in A] = |\{A \in \mathcal{F} : i \in A\}|/|\mathcal{F}|$. Therefore Eq. (31) translates to

$$\mathbb{P}[i \in A] \;<\; 0.01 \qquad \text{for every } i \in [n]. \tag{32}$$

In particular, $\mathbb{P}[i \in A] \leq 0.01$ holds for every $i$, and the hypotheses of Theorem 4.1 are satisfied.

**Step 4: $\mathrm{H}(A) > 0$ from $|\mathcal{F}| \geq 2$.** Since $|\mathcal{F}| \geq 2$ and $A$ is uniform on $\mathcal{F}$,

$$\mathrm{H}(A) \;=\; \log|\mathcal{F}| \;\geq\; \log 2 \;=\; 1 \;>\; 0. \tag{33}$$

**Step 5: Derive the contradiction.** Applying Theorem 4.1,

$$\mathrm{H}(A \cup B) \;\geq\; 1.26\,\mathrm{H}(A). \tag{34}$$

Combining Eq. (34) with Eq. (30),

$$1.26\,\mathrm{H}(A) \;\leq\; \mathrm{H}(A \cup B) \;\leq\; \mathrm{H}(A),$$

which forces $0.26\,\mathrm{H}(A) \leq 0$, i.e. $\mathrm{H}(A) \leq 0$. This contradicts Eq. (33).

**Conclusion.** The supposition Eq. (31) fails for at least one $i \in [n]$; that is, some $i \in [n]$ satisfies $|\{A \in \mathcal{F} : i \in A\}| \geq 0.01 \cdot |\mathcal{F}|$. Taking $c = 0.01$ gives Eq. (28). This completes the proof. □

*Remark* 5.2. The constant $c = 0.01$ in Theorem 5.1 is not tight. Gilmer's original paper [3] already notes that the constant was not optimized and that the same proof technique, with a more careful analysis of the constants in Lemma 3.1, would yield a strictly larger admissible $c$. Subsequent work (Sawin; Chase-Lovett; Alweiss-Huang-Sellke; all in late 2022, within days of Gilmer's preprint) showed that the entropy approach can be pushed to $c \to (3 - \sqrt{5})/2 \approx 0.38$, which is the maximum achievable by this technique without further input. The Frankl conjecture [2], which asks whether $c = 1/2$ is admissible, remains open.

## References


[1] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory.* John Wiley & Sons, 2 edition, 2006.

[2] Péter Frankl. Extremal set systems. *Handbook of Combinatorics*, 2:1293–1329, 1995.

[3] Justin Gilmer. A constant lower bound for the union-closed sets conjecture. *arXiv preprint arXiv:2211.09055*, 2022. URL `https://arxiv.org/abs/2211.09055`. Version v2 (28 Nov 2022).